```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips=sns.load_dataset('tips')
```

```
flights=sns.load_dataset('flights')
```

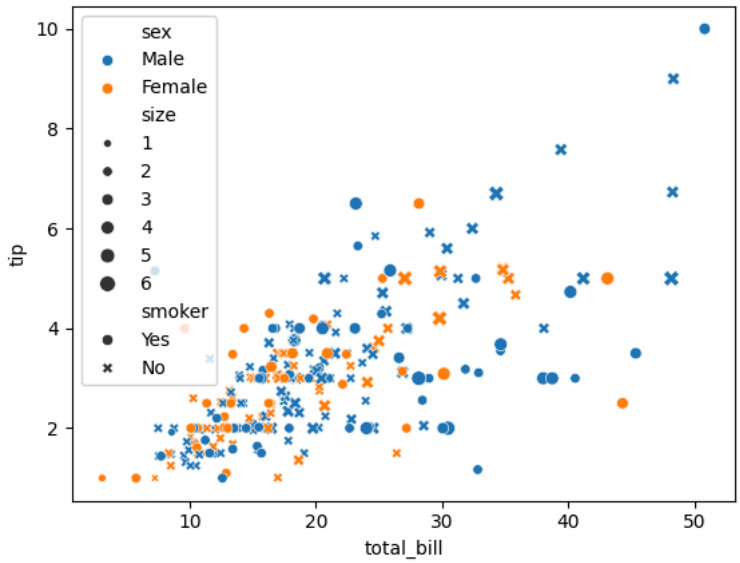
```
iris=sns.load_dataset('iris')
```

```
titanic=pd.read_csv('titanic.csv')
```

```
#1.scatterplot(numerical-numerical)

#sns.scatterplot(data=tips,x='total_bill',y='tip')
sns.scatterplot(x=tips['total_bill'],y=tips['tip'],hue=tips['sex'],style=tips['smoker'],size=tips['size'])
```

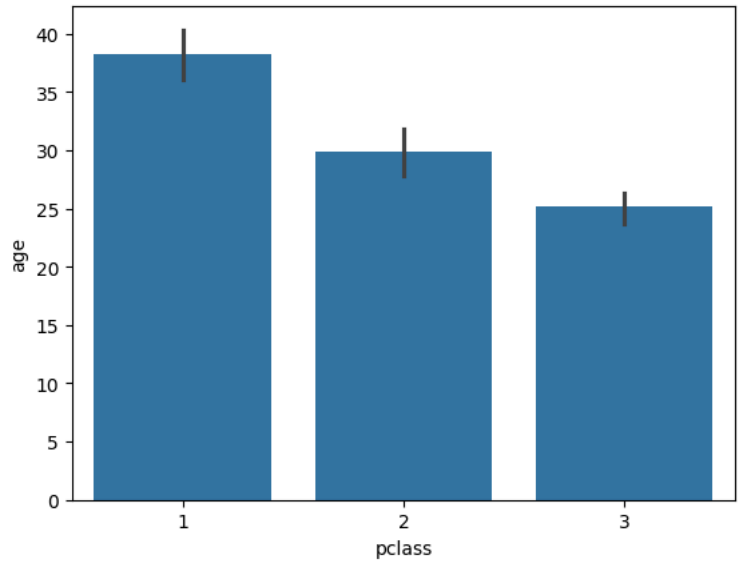
```
<Axes: xlabel='total_bill', ylabel='tip'>
```




```
#2.barplot(numerical-categorical)
#pclass ke according avg age kya hai travelers ki

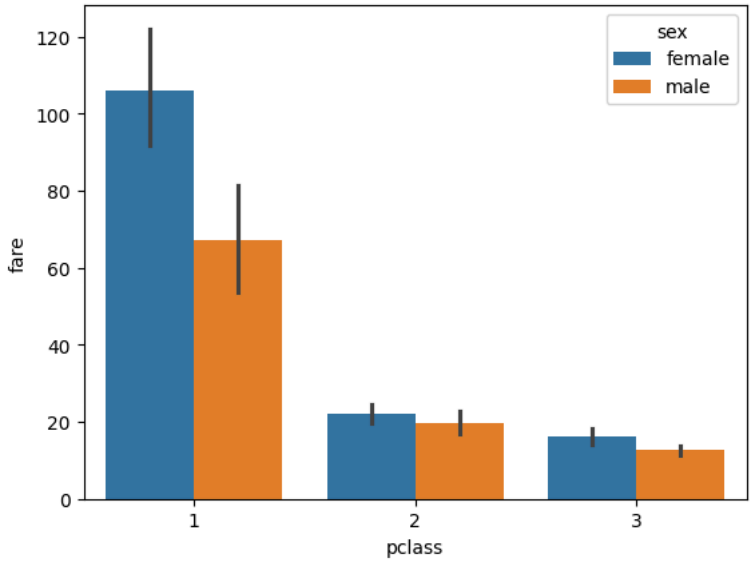
sns.barplot(data=titanic,x='pclass',y='age')
```

```
<Axes: xlabel='pclass', ylabel='age'>
```




```
sns.barplot(data=titanic,x='pclass',y='fare',hue='sex')
```

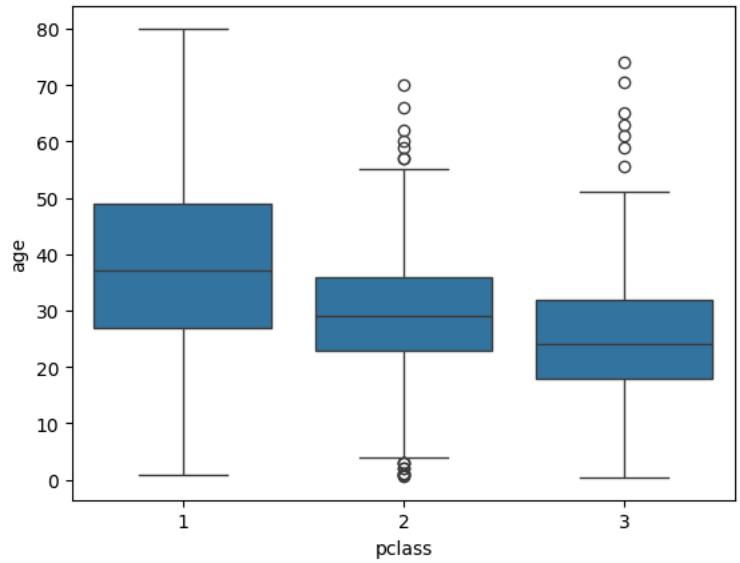
```
<Axes: xlabel='pclass', ylabel='fare'>
```




```
# 3.boxplot(numerical-categorical)

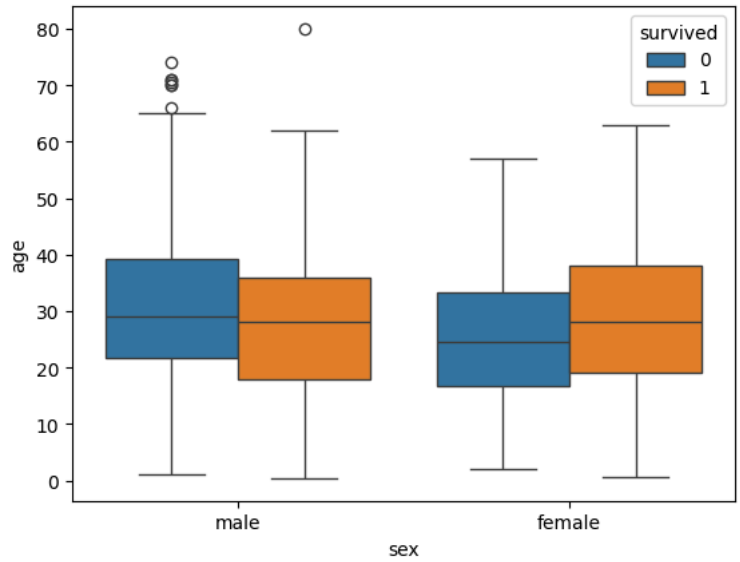
sns.boxplot(data=titanic,x='pclass',y='age')
```

```
<Axes: xlabel='pclass', ylabel='age'>
```




```
sns.boxplot(data=titanic,x='sex',y='age',hue='survived')
```

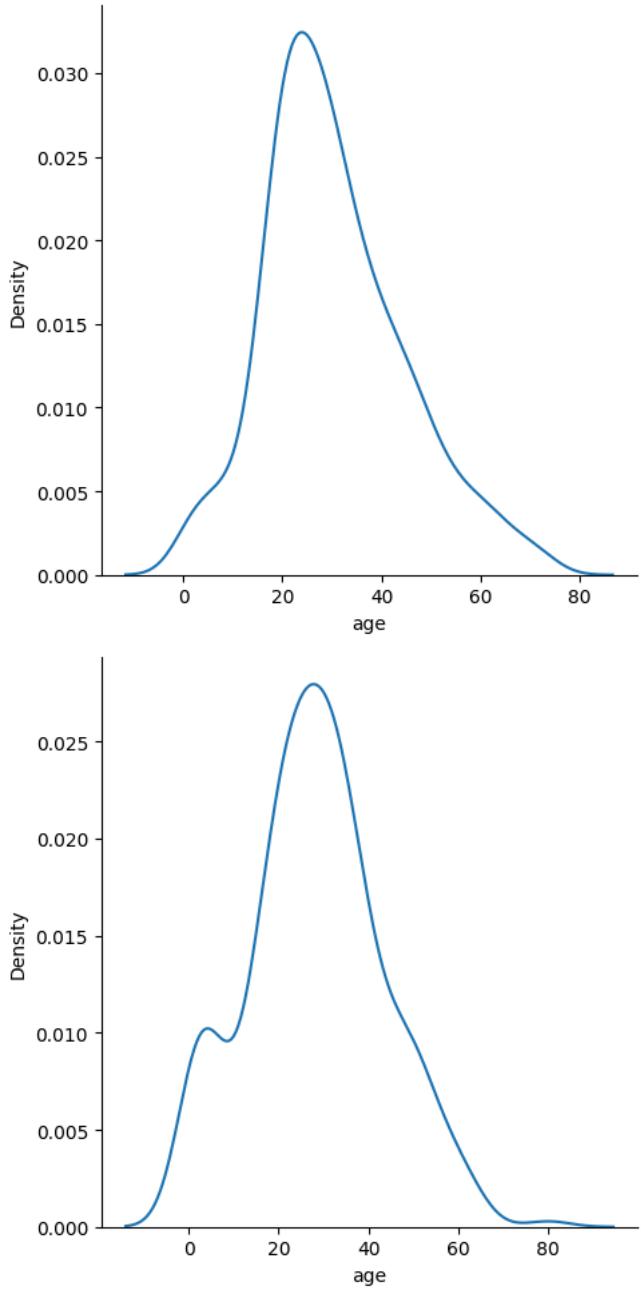
```
<Axes: xlabel='sex', ylabel='age'>
```



```
# 4.distplot(numerical-categorical)

sns.displot(titanic[titanic['survived']==0]['age'],kind='kde')
sns.displot(titanic[titanic['survived']==1]['age'],kind='kde')  #dono code ke graph ko ek he graph mai merge krne ke liye r
```

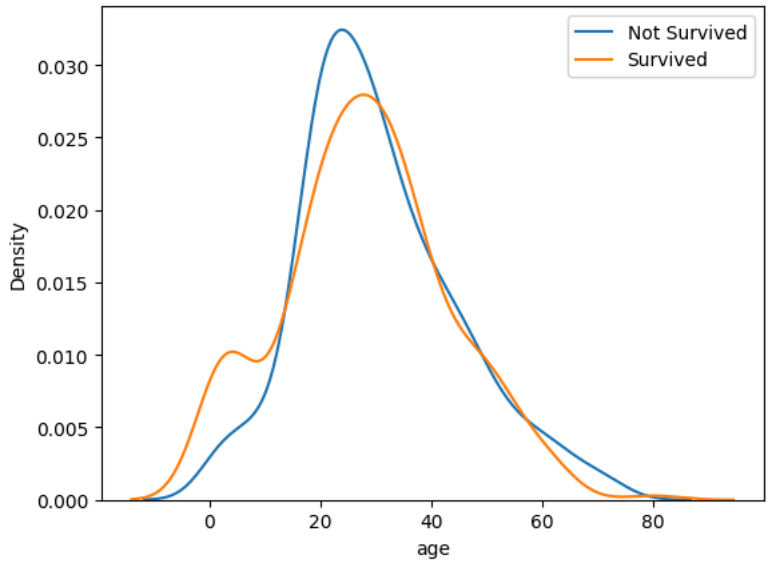
```
<seaborn.axisgrid.FacetGrid at 0x7ac0413f9950>
```




```
sns.kdeplot(
    titanic[titanic['survived'] == 0]['age'],
    label='Not Survived'
)

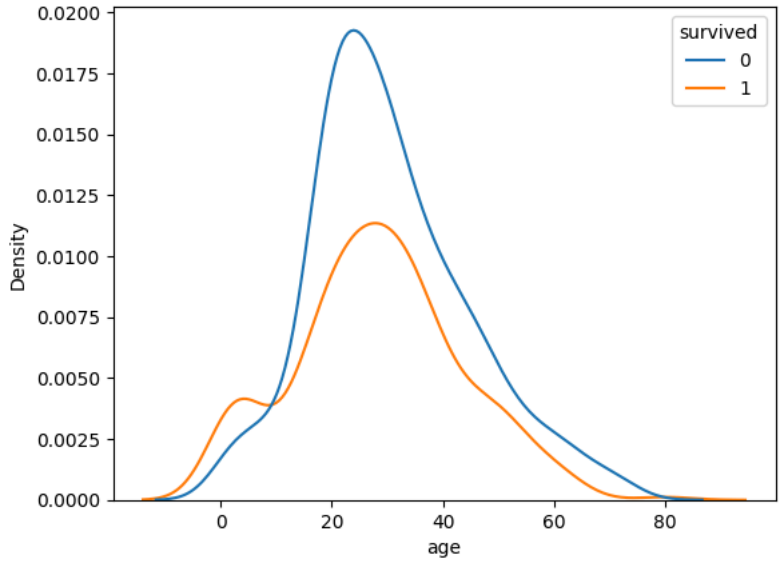
sns.kdeplot(
    titanic[titanic['survived'] == 1]['age'],
    label='Survived'
)

plt.legend()
plt.show()   #uppar vale dono code ke graph ko ek he graph mai merge krne ke liye
```




```
sns.kdeplot(
    data=titanic,
    x='age',
    hue='survived'
)               #uppar vale dono code ke graph ko ek he graph mai merge krne ke liye
```

```
<Axes: xlabel='age', ylabel='Density'>
```




```
# 5.heatmap(categorical categorical)

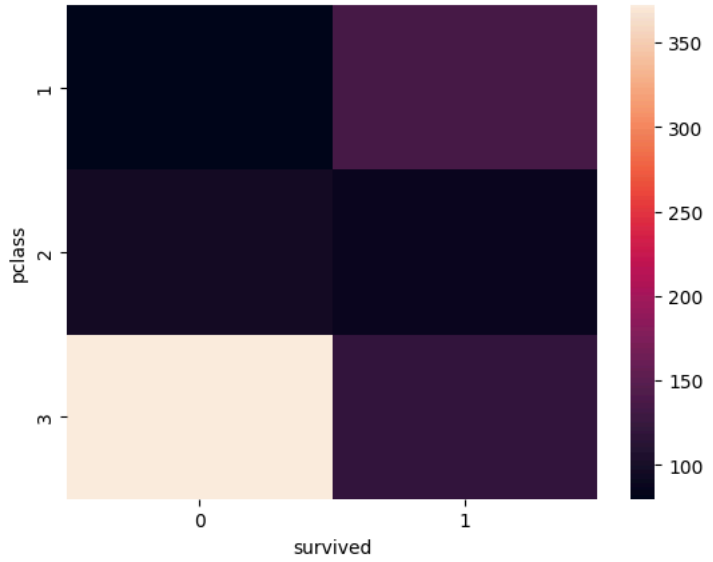
# do categorical columns pclass or survived mens kis class mai kitne log bache or kitne mare

pd.crosstab(titanic['pclass'],titanic['survived'])
```

survived	0	1
pclass		
1	80	136
2	97	87
3	372	119

```
#is data ko map mai dekhne ke liye ==>
sns.heatmap(pd.crosstab(titanic['pclass'],titanic['survived']))
```

```
<Axes: xlabel='survived', ylabel='pclass'>
```



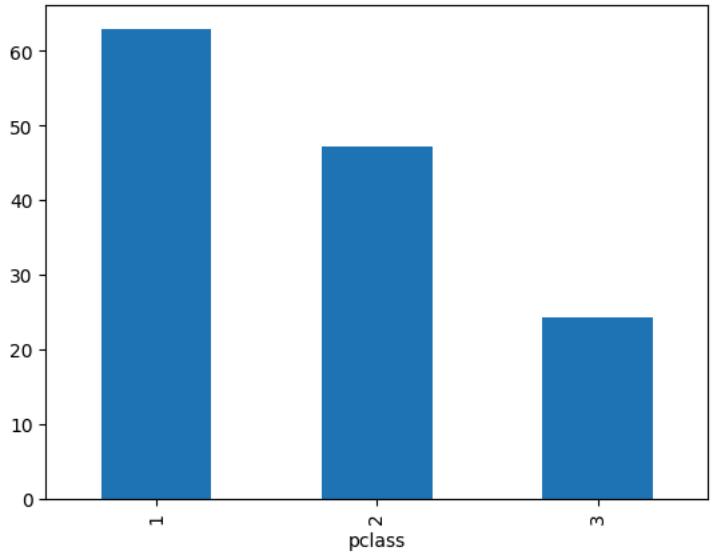

```
#or % mai dekhne ke liye grph hoga==>
titanic.groupby('pclass')['survived'].mean()*100
```

pclass	survived
1	62.962963
2	47.282609
3	24.236253

dtype: float64

```
#graph ke liye==>
(titanic.groupby('pclass')['survived'].mean()*100).plot(kind='bar')
```

```
<Axes: xlabel='pclass'>
```




```
titanic.groupby('sex')['survived'].mean()*100
```

sex	survived
female	74.203822
male	18.890815

dtype: float64

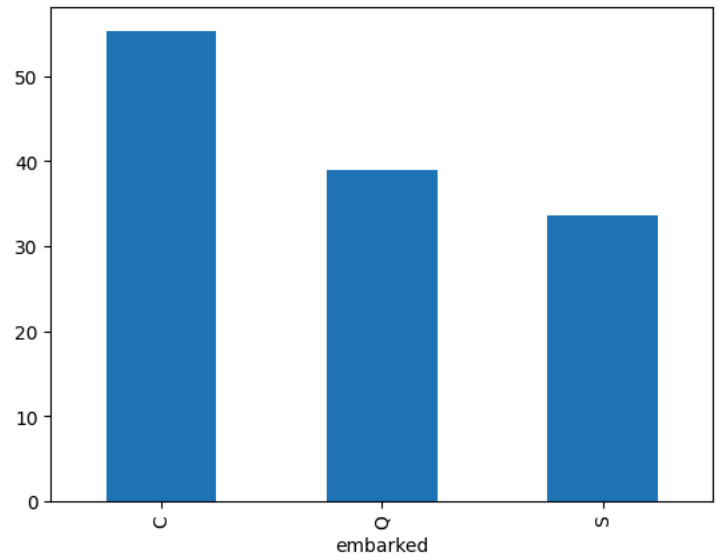
```
titanic.groupby('embarked')['survived'].mean()*100
```

embarked	survived
C	55.357143
Q	38.961039
S	33.695652

dtype: float64

```
(titanic.groupby('embarked')['survived'].mean()*100).plot(kind='bar')
```

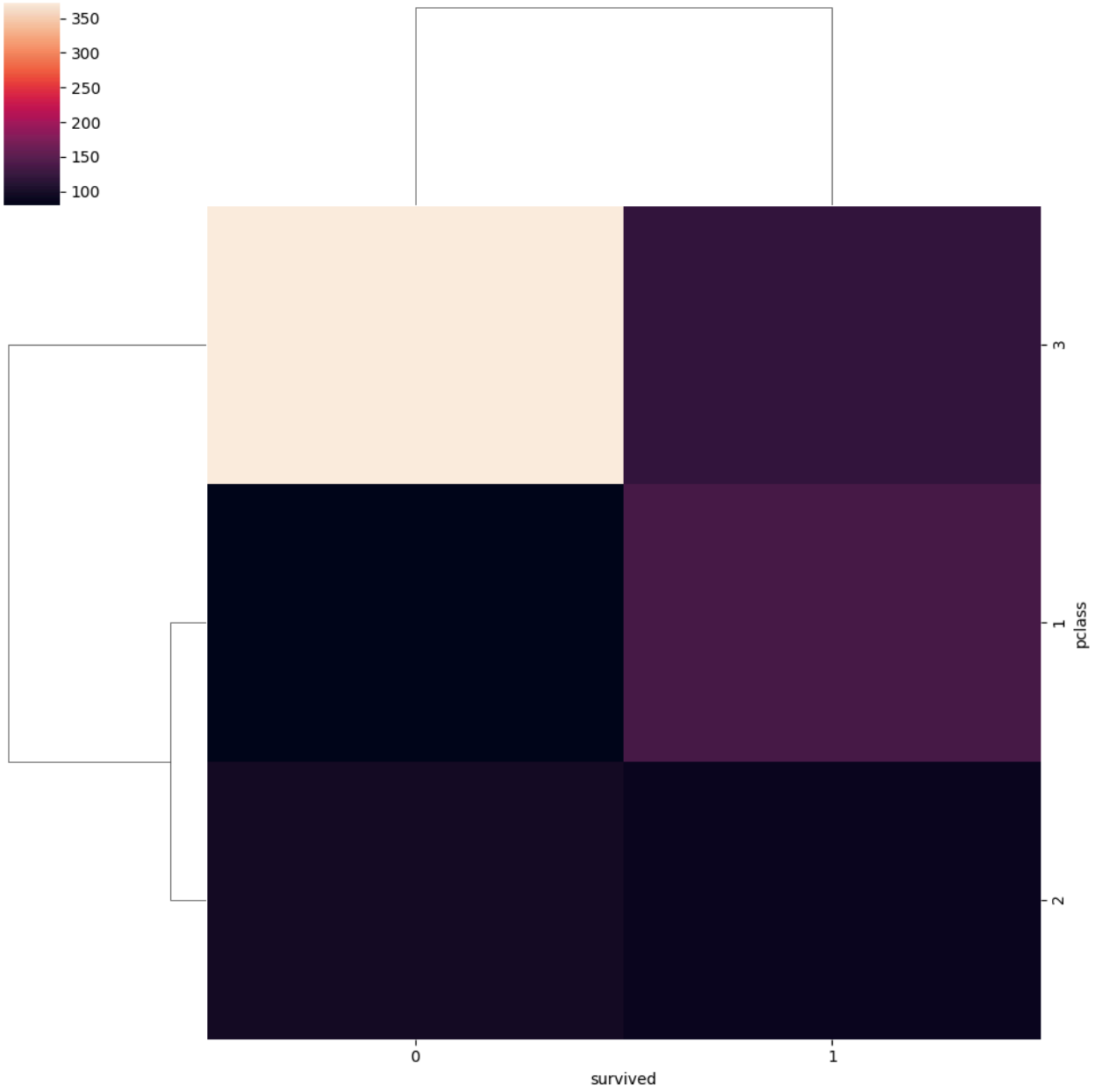
```
<Axes: xlabel='embarked'>
```




```
# 6.clustermap(categorical-categorical)
# ye values ke beech ka behave or closness show krta h graph mai color ke behalf

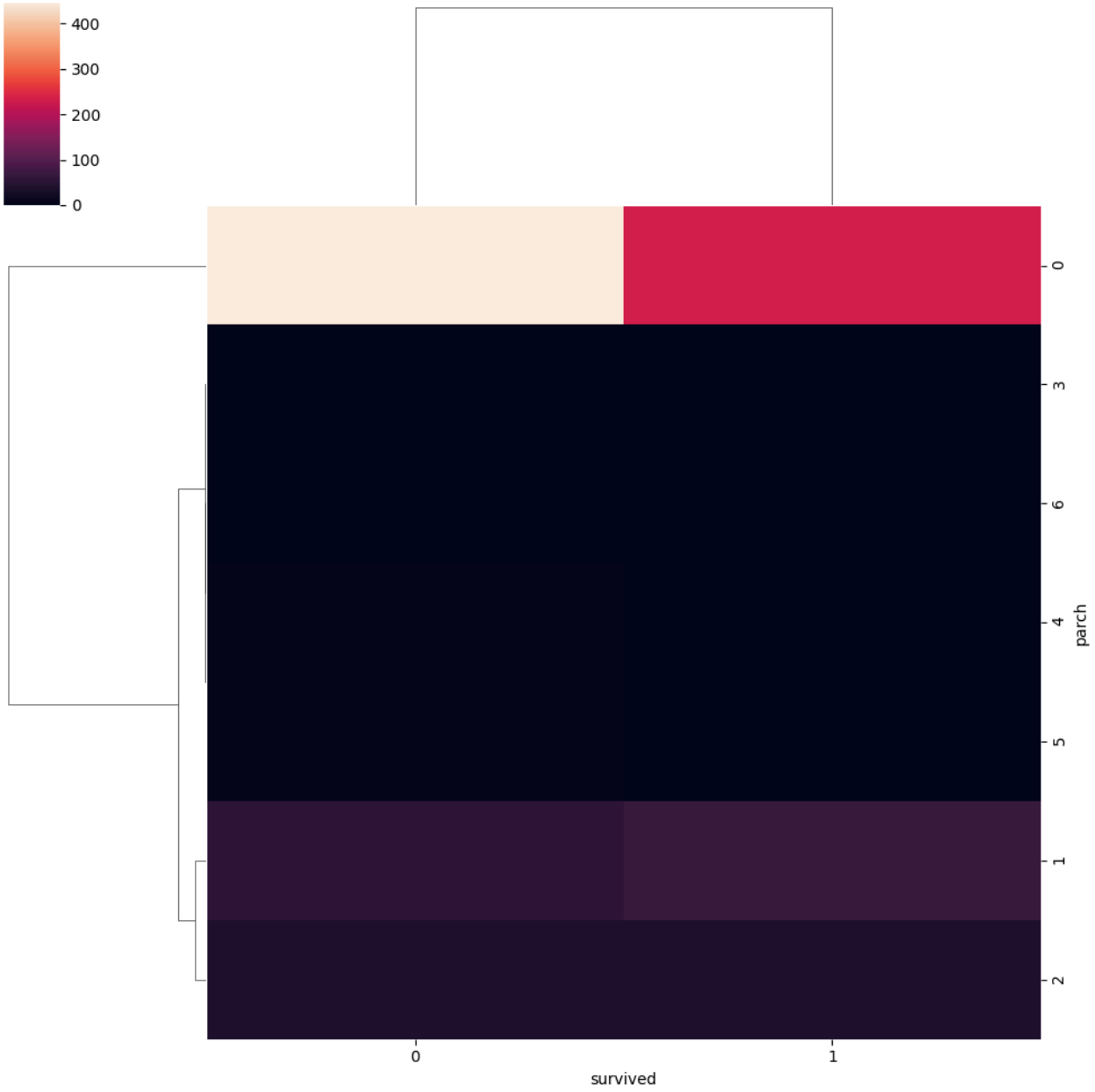
sns.clustermap(pd.crosstab(titanic['pclass'],titanic['survived']))
```

```
<seaborn.matrix.ClusterGrid at 0x7ac040ec7890>
```




```
sns.clustermap(pd.crosstab(titanic['parch'],titanic['survived']))
```

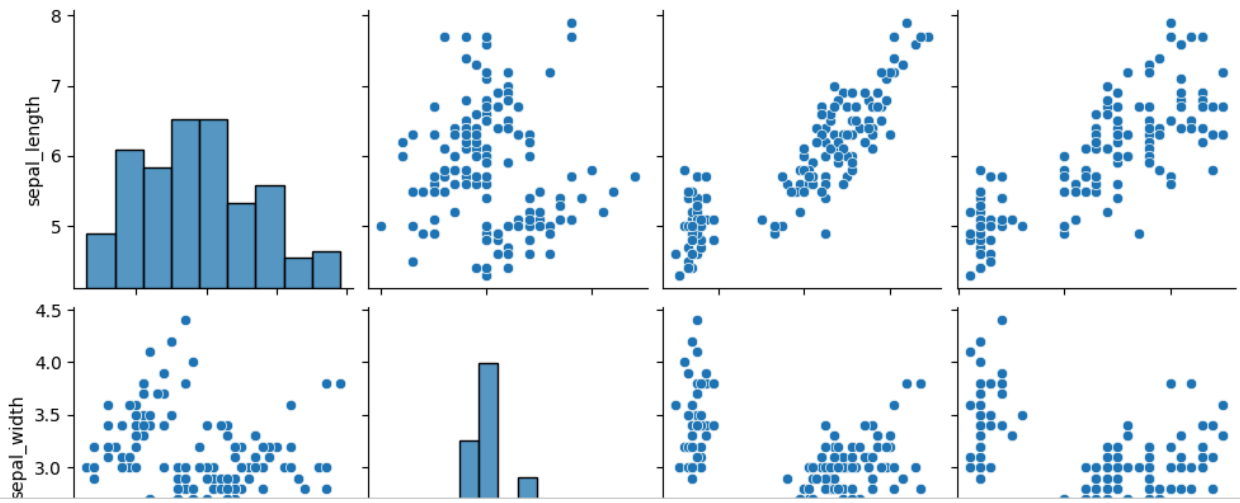
```
<seaborn.matrix.ClusterGrid at 0x7ac040ddd950>
```




```
#7.pairplot


sns.pairplot(iris)
sns.pairplot(iris,hue='species')
```

```
<seaborn.axisgrid.PairGrid at 0x7ac05360ed50>
```

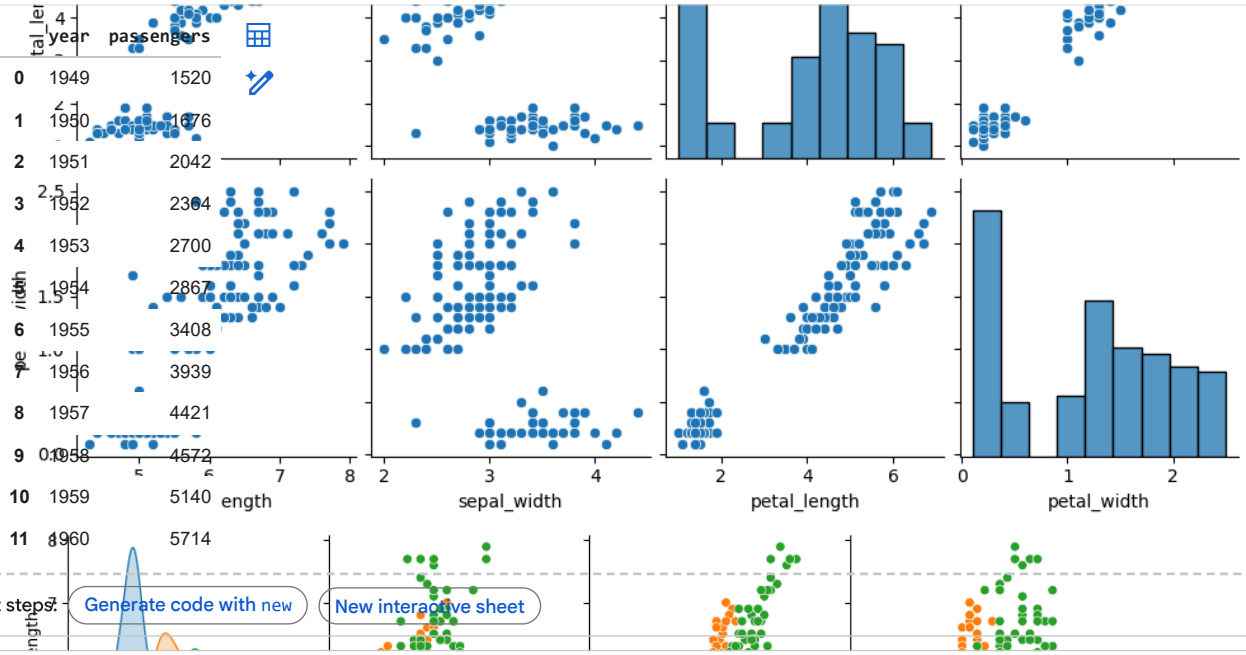



```
#8.lineplot(numerical-numerical)
#use jab accha h jab x-axis mai quantity time based ho like month,year..or jo aage bhad rahi ho

new=flights.groupby('year')['passengers'].sum().reset_index()
new
```

	year	passengers
0	1949	1520
1	1950	1676
2	1951	2042
3	1952	2364
4	1953	2700
5	1954	2867
6	1955	3408
7	1956	3939
8	1957	4421
9	1958	4572
10	1959	5140
11	1960	5714

Next steps: Generate code with new | New interactive sheet




```
sns.lineplot(data=new,x='year',y='passengers')
```

```
<Axes: xlabel='year', ylabel='passengers'>
```

